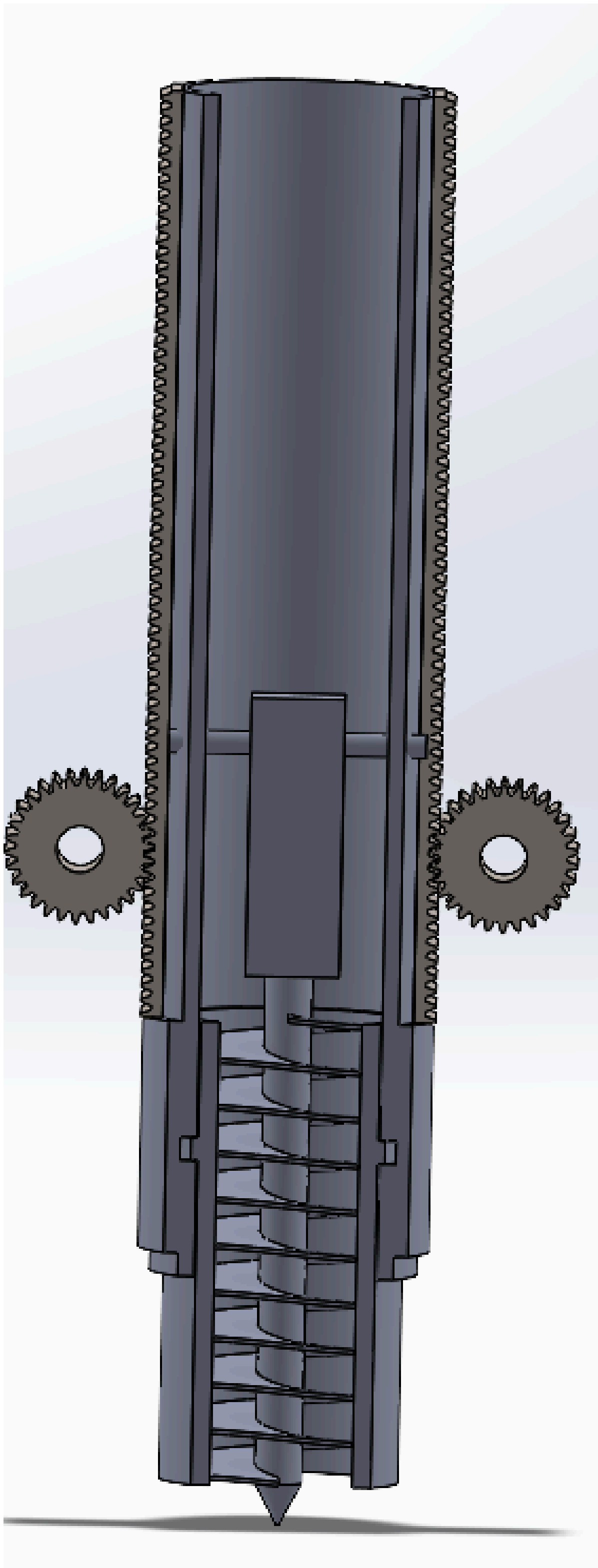
During my sophomore fall, I had decided that I was going to declare my major to be Mechanical Engineering and so I started looking for extracurricular activities to participate in to try and get some hands-on experience and apply some of the principles I was beginning to learn about in class. It was also around this time that a classmate was forming the Harvard Undergraduate Robotics Club, or HURC. I was lucky enough to sit in on the first few meetings of the growing membership, where all we knew was that we needed to build a rover for a Mars rover competition, and literally nothing else. What seemed like a bunch of undergrads who didn't really know anything about engineering trying to build something super complex on an unreasonable time scale (because it was) was also the perfect opportunity for me to start to familiarize myself with the engineering world, take ownership of problems, and be creative.

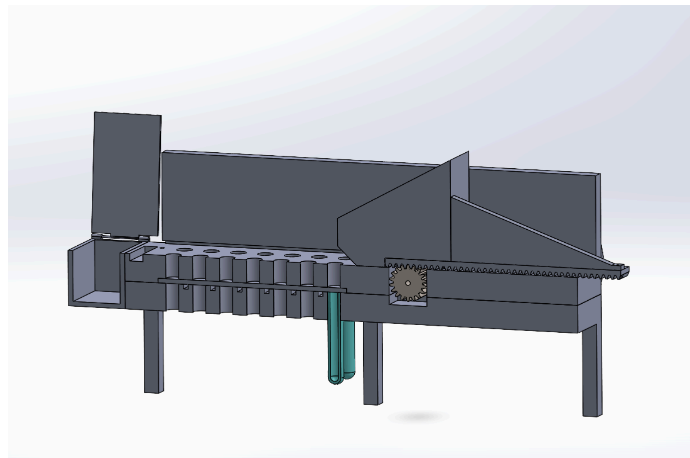
On this early team I gave input on early design decisions, like chassis and suspension designs, but my role quickly became more focused on designing the dirt-sample collection and testing mechanism. As part of this rover competition, the rover needed to be able to collect a dirt sample, and mix it with various chemicals to analyze and detect organic matter, etc. I came up with, and drafted in CAD a retractable Archimedes' screw which would drill into the soil and collect the sample. I also designed a collection table with a dirt compacting mechanism to appropriately measure and distribute the samples to easily-detachable test tubes preloaded with chemicals. This later transitioned into a dual purpose centrifuge/soil-distributer design.

Along with exercising my design and CAD muscles, the club also motivated me to become more familiar with fabrication techniques. I became certified in basic shop training, milling, and welding to support the club. As the club was so new, there was a long time where I was one of only a few people trained and certified to use the machine shop equipment, and as such I carried out many manufacturing duties including milling, welding, drilling, tapping, etc. I spent many hours in the machine shop, and I feel that the opportunity I had to see engineering from both a design, analysis, and manufacturing standpoint has been an important part of my education.

Below, you'll find a gallery of photos illustrating my work with HURC, along with short descriptions of what they show:



On the left is the Archimedes' screw dirt collector. I designed it to be made of PVC tubing and an auger bit from the Home Depot. The rack and pinion gear system would drive it down into the soil for extraction.



Above is the original dirt collection mechanism. The dirt would deposit onto the tray and be pushed and compressed into the intentionally sized holes, and off the tray into the extra collection bin. When the flat middle bit was pulled out, the dirt would fall into the screwed in test tubes, and mix for testing.

Below is a later iteration adapted to sit on a conical centrifuge. Here, the dirt would fall in the holes which would be open when the mechanism turns right, but closed when it turned left, so the test tubes could still be opened and closed, and now spun.

On the left is a part that I was conversationally milling. I programmed the circular profile into the mill and ran it after setting the origin and drilling the four other holes seen. I had to make a few of these, which I think were used on the suspension system.

Below is an early version of the robotic arm the rover has. I helped manufacture many of the components and assemble them. I also spent some time over the summer positioning designing mounts for limit switches for this arm.